

FEB COMPOSITES

The SN6 workspace — what it does, how it's built, and what I need from you

[Your name] · [Meeting date]

Live at feb-composites.web.app

- 1 What SN5 cost
- 2 What it is
- 3 One part, end to end
- 4 The traveler
- 5 Running the season
- 6 Why it survives handoff
- 7 Limits and the ask

Three from the SN5 review

PP-01	Duratec	Days of mold time per incident, re-coating after sand-through.
PP-02	Customs	A ~109 lb Easy Composites order sat with no ETA. Rush replacements elsewhere: \$120 of ACP rods, ~\$400 to expedite.
PP-09	“What stack did we use?”	Asked in March about last year's seat. The tracker recorded intent, not what was actually laid.

10 pain points, root-caused, each mapped to a standard that fixes it

14 numbered standards, CS-000 through CS-013. CS-INDEX is the lookup.

`tools/check_traceability.py` audits that every pain point still points at a live standard.

Each one lands in the app as a step on a work order.

FEB Composites

Dashboard

Work Orders

Parts

Stock

Tickets

Timeline

Weekly Plan

Budget

Documents

Reports

People

Dashboard

2YOUR OPEN ITEMS

5BEHIND SCHEDULE

\$2054SEASON SPEND

YOUR OPEN ITEMS (2)

ITEM	WHO	DEADLINE
TICKET Order carbon before the customs window closes	Nick Jepsen	2026-08-04 (4d)
TICKET Move Firebase project to a team account	Nick Jepsen	2026-08-08 (8d)

BEHIND SCHEDULE (5)

ITEM	WHO	DEADLINE
PART DASHBOARD	Justin	2026-01-25 (187d late)
PART UT SIDE LEFT	Justin / Justin	2026-03-29 (124d late)
PART UT SIDE RIGHT	Justin / Justin	2026-04-05 (117d late)
PART UT DIFFUSER	Justin / Justin	2026-04-05 (117d late)
TICKET Drop-test rig needs a new load cell	Theo Brooks	2026-07-24 (7d late)

UPCOMING TEAM DEADLINES (NEXT 14 DAYS)

ITEM	WHO	DEADLINE
TICKET Order carbon before the customs window closes	Nick Jepsen	2026-08-04 (4d)
TICKET Qualify XCR as the standing mold sealer	Ana Rivera	2026-08-07 (7d)
TICKET Move Firebase project to a team account	Nick Jepsen	2026-08-08 (8d)
TICKET Infusion line kinks at the nosecone corner	David Okafor	2026-08-11 (11d)
TICKET Vacuum pump loses 3 inHg overnight	Jai Patel	2026-08-12 (12d)
TICKET Field-verify the CS-011 storage map at RFS	Simon Starbuck	2026-08-14 (14d)

02 WHAT IT IS

One shared workspace, live for everyone at once

Free to run: Firebase Blaze with a \$1–5 billing cap as the backstop.

Figures on these screens are seeded demo data, not real spend.

An account gets you nothing until a lead adds your email

Enforced in firestore.rules, server-side
— not by hiding buttons.

Roster self-edits are limited to name
and avatar, so nobody can promote
themselves to lead.

A member does the day-to-day work. A
lead can also delete, restore from
backup, and manage the roster.

The screenshot shows the 'FEB Composites' application interface. On the left is a dark blue sidebar with a menu: Dashboard, Work Orders, Parts, Stock, Tickets, Timeline, Weekly Plan, Budget, Documents, Reports, and People (highlighted). The main content area is titled 'People' and shows a list of 8 people on the roster. At the top right of the main area are buttons for Backup, Restore, Load SN5 archive, Roster, and a user profile for Nick Jepsen (lead) with a Sign out button. Below the title bar is a search bar and the text '8 people on the roster'. The roster is a table with columns: PERSON, ROLE, and ASSIGNMENTS.

PERSON	ROLE	ASSIGNMENTS
Ana Rivera a.rivera@berkeley.edu	member	Qualify XCR as the standing mold sealer WO-SN6-004
David Okafor d.okafor@berkeley.edu	member	Infusion line kinks at the nosecone corner WO-SN6-004
Jai Patel j.patel@berkeley.edu	member	Vacuum pump loses 3 inHg overnight
Lena Novak l.novak@berkeley.edu	member	Sponsor outreach: Sigmatex fabric donation
Mei Chen m.chen@berkeley.edu	member	Write the layup induction for new members
Nick Jepsen (you) n.jepsen@berkeley.edu	lead	Order carbon before the customs window closes Move Firebase project to a team account
Simon Starbuck simon.starbuck@berkeley.edu	member	Field-verify the CS-011 storage map at RFS
Theo Brooks t.brooks@berkeley.edu	member	Drop-test rig needs a new load cell

Eleven tabs. The rest of this is four of them.

Cross-links everywhere — click a chip to jump to the related record. ⌘K for global search.

- **Work Orders**
- **Parts**
- **Stock**
- **Budget**

Dashboard

Tickets

Timeline

Weekly Plan

Documents

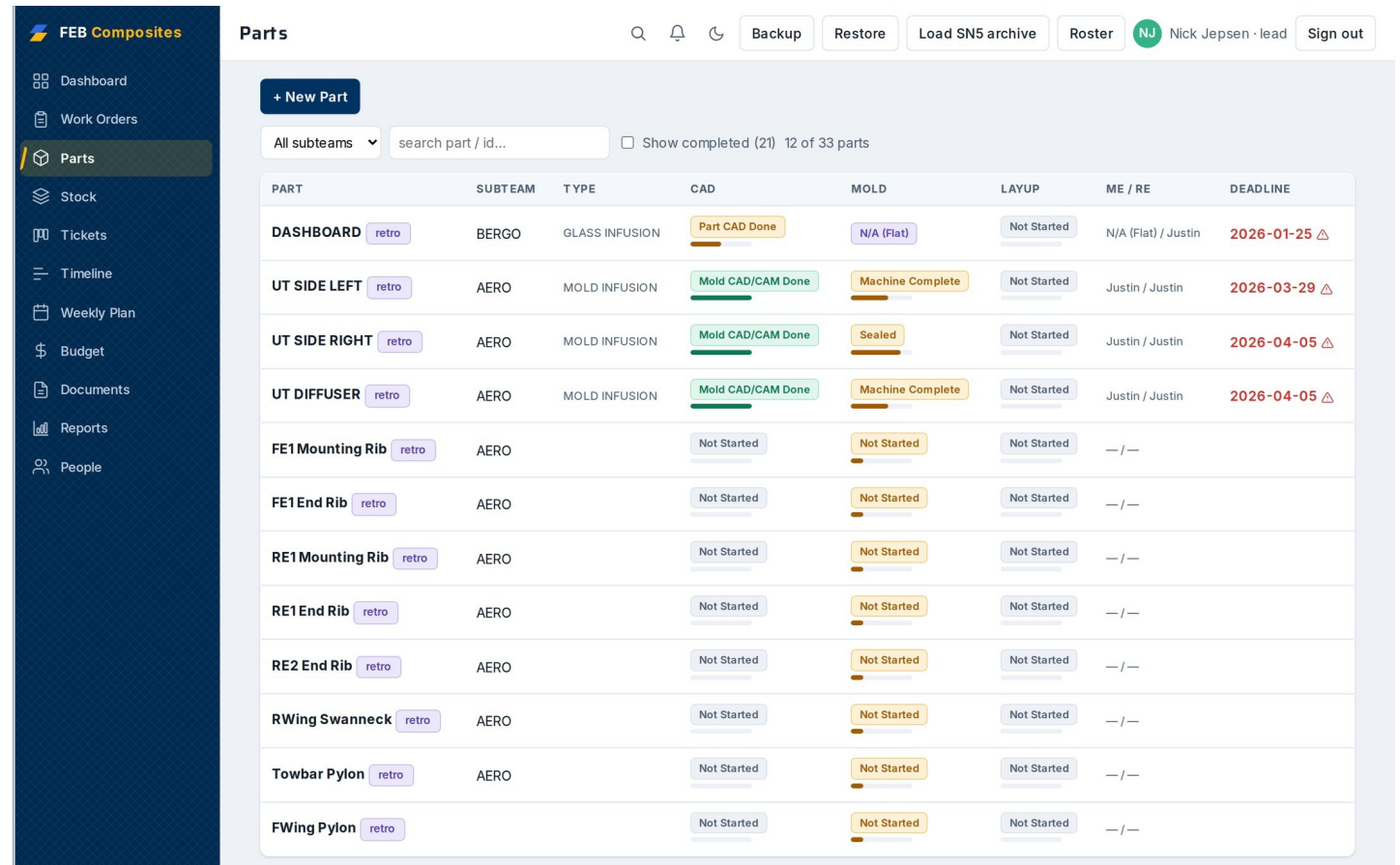
Reports

People

CAD, mold and layup move independently, so the tracker shows three bars, not one percentage

Subteam, engineers, target weight, layup deadline.

A part's layup stack and its work order's stack stay in sync — edit either.



FEB Composites

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People

Parts

+ New Part

All subteams ▼ search part / id... ☐ Show completed (21) 12 of 33 parts

PART	SUBTEAM	TYPE	CAD	MOLD	LAYUP	ME / RE	DEADLINE
DASHBOARD retro	BERGO	GLASS INFUSION	Part CAD Done	N/A (Flat)	Not Started	N/A (Flat) / Justin	2026-01-25 ⚠
UT SIDE LEFT retro	AERO	MOLD INFUSION	Mold CAD/CAM Done	Machine Complete	Not Started	Justin / Justin	2026-03-29 ⚠
UT SIDE RIGHT retro	AERO	MOLD INFUSION	Mold CAD/CAM Done	Sealed	Not Started	Justin / Justin	2026-04-05 ⚠
UT DIFFUSER retro	AERO	MOLD INFUSION	Mold CAD/CAM Done	Machine Complete	Not Started	Justin / Justin	2026-04-05 ⚠
FE1 Mounting Rib retro	AERO		Not Started	Not Started	Not Started	— / —	
FE1 End Rib retro	AERO		Not Started	Not Started	Not Started	— / —	
RE1 Mounting Rib retro	AERO		Not Started	Not Started	Not Started	— / —	
RE1 End Rib retro	AERO		Not Started	Not Started	Not Started	— / —	
RE2 End Rib retro	AERO		Not Started	Not Started	Not Started	— / —	
RWing Swanneck retro	AERO		Not Started	Not Started	Not Started	— / —	
Towbar Pylon retro	AERO		Not Started	Not Started	Not Started	— / —	
FWing Pylon retro			Not Started	Not Started	Not Started	— / —	

A full sheet and an offcut are the same kind of record

So remnants come back into stock instead of piling up under the bench.

The planner picks thicknesses from what is actually on the rack — there is no point offering a stack we don't own.

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Stock

Q

B

C

Backup

Restore

Load SN5 archive

Roster

NJ Nick Jepsen · lead

Sign out

+ Add board

Plan a mold

All stock

search label / id...

8 of 8 boards

ON HAND

38.1MM · 30 LB

50.8MM · 30 LB

50.8MM · 60 LB

76.2MM · 30 LB

6.75 m²

9.91 m²

2.97 m²

6.14 m²

BOARD	LENGTH	WIDTH	THICKNESS	DENSITY	QTY	KIND	
rack A, middle	96"	48"	1.5"	30 lb/ft ³	2	sheet	
rack B, leaning · from NOSECONE	33"	19"	1.5"	30 lb/ft ³	2	offcut	
rack A, top shelf	96"	48"	2"	30 lb/ft ³	3	sheet	
rack B — 60 pcf, seals better	96"	48"	2"	60 lb/ft ³	1	sheet	
rack B, leaning · from UT SIDE LEFT	46"	30"	2"	30 lb/ft ³	1	offcut	
under the bench, smallest worth keeping · from UT DIFFUSER	10"	4"	2"	30 lb/ft ³	4	offcut	
rack A, bottom	96"	48"	3"	30 lb/ft ³	2	sheet	
under the bench · from CLAMSHELL	22"	14"	3"	30 lb/ft ³	1	offcut	

A real Fusion export is an assembly. It asks which body you mean.

Rather than slicing the bounding box of everything and planning a void.

Three sample molds ship with the app, so the planner works before anyone exports anything.

Plan a mold

NAME
Clamshell — upper

START FROM
an STL file — beta

MOLD STL
Choose File No file chosen

...OR TRY A SAMPLE
Clamshell assembly — 3 separate bodies in one file

Shipped with the app, so you can see the planner work without exporting anything first.

STL UNITS
millimetres

An STL carries no units. Getting this wrong is a 25.4 × mistake.

Beta. Real exports still turn up surprises — assemblies holding many bodies, rough meshes, odd draft. Check the stack view before anyone cuts, and fall back to dimensions if it looks wrong.

BOARDS
choose them for me, from stock

Available on the rack: 1.5", 2", 3"

WHICH BODY?
#1 — 14.0 × 10.0 × 3.2 in (2,848 tris)

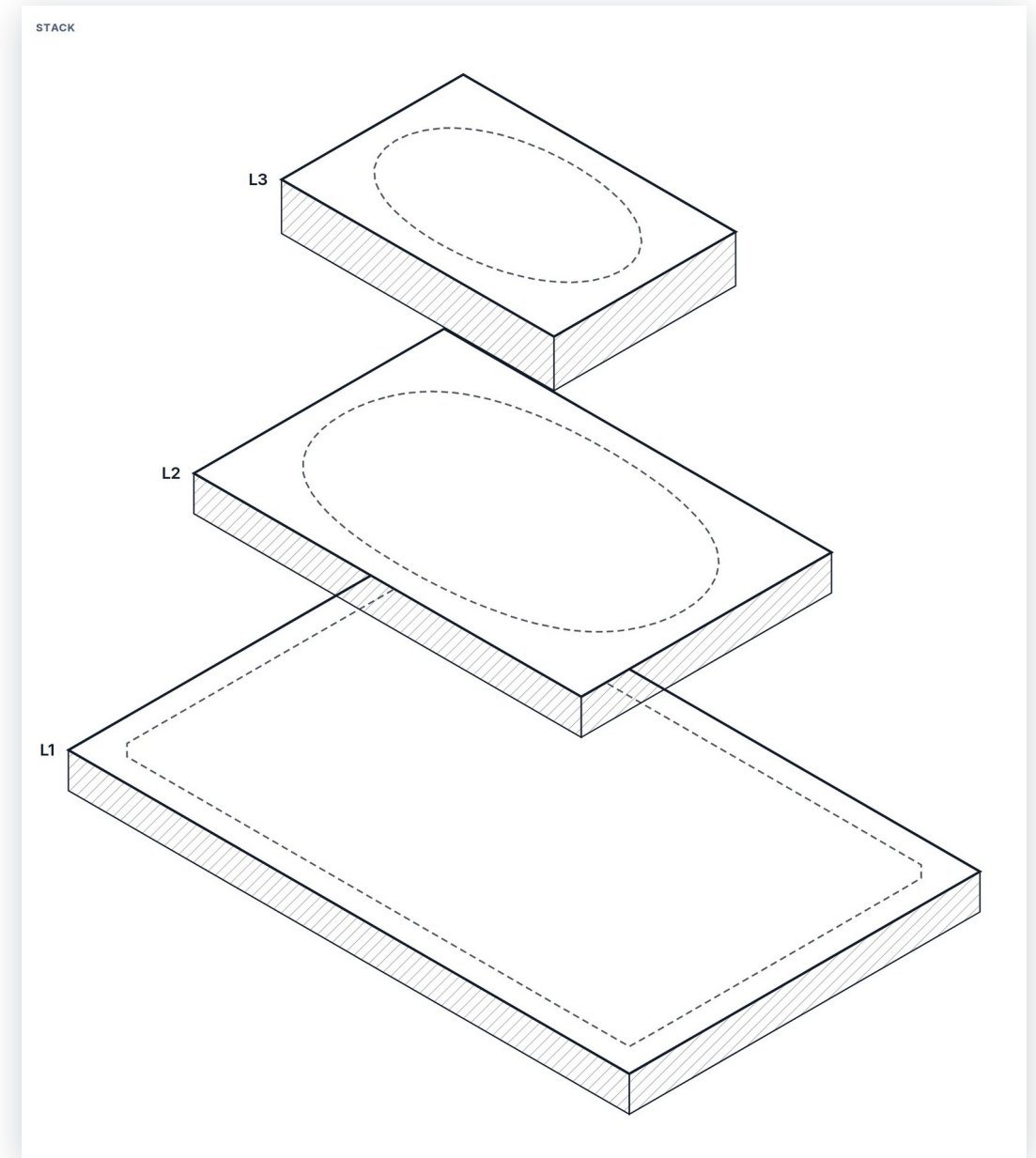
This file holds 3 separate bodies — an assembly export, not one mold. Plan them one at a time.

Cancel Plan

Hand it a mold. It works out which boards to glue and how to saw them.

Thicknesses chosen to waste the least board, from the rack you actually have.

Tall molds split at the ShopSabre's ~6in cut depth, into one machine setup per section.



BLANKS TO CUT

LAYER	THICKNESS	BLOCKS	BLANK SIZE	DATUM OFFSET
L1	38.1 mm (1.5")	1	609.6 mm (24") × 381 mm (15")	X -25.4 mm (-1"), Y -25.4 mm (-1")
L2	38.1 mm (1.5")	1	421 mm (16.57") × 271.9 mm (10.71")	X 56.5 mm (2.22"), Y 29.1 mm (1.15")
L3	50.8 mm (2")	1	296.1 mm (11.66") × 197.5 mm (7.78")	X 114.5 mm (4.51"), Y 66.3 mm (2.61")

03 ONE PART, END TO END

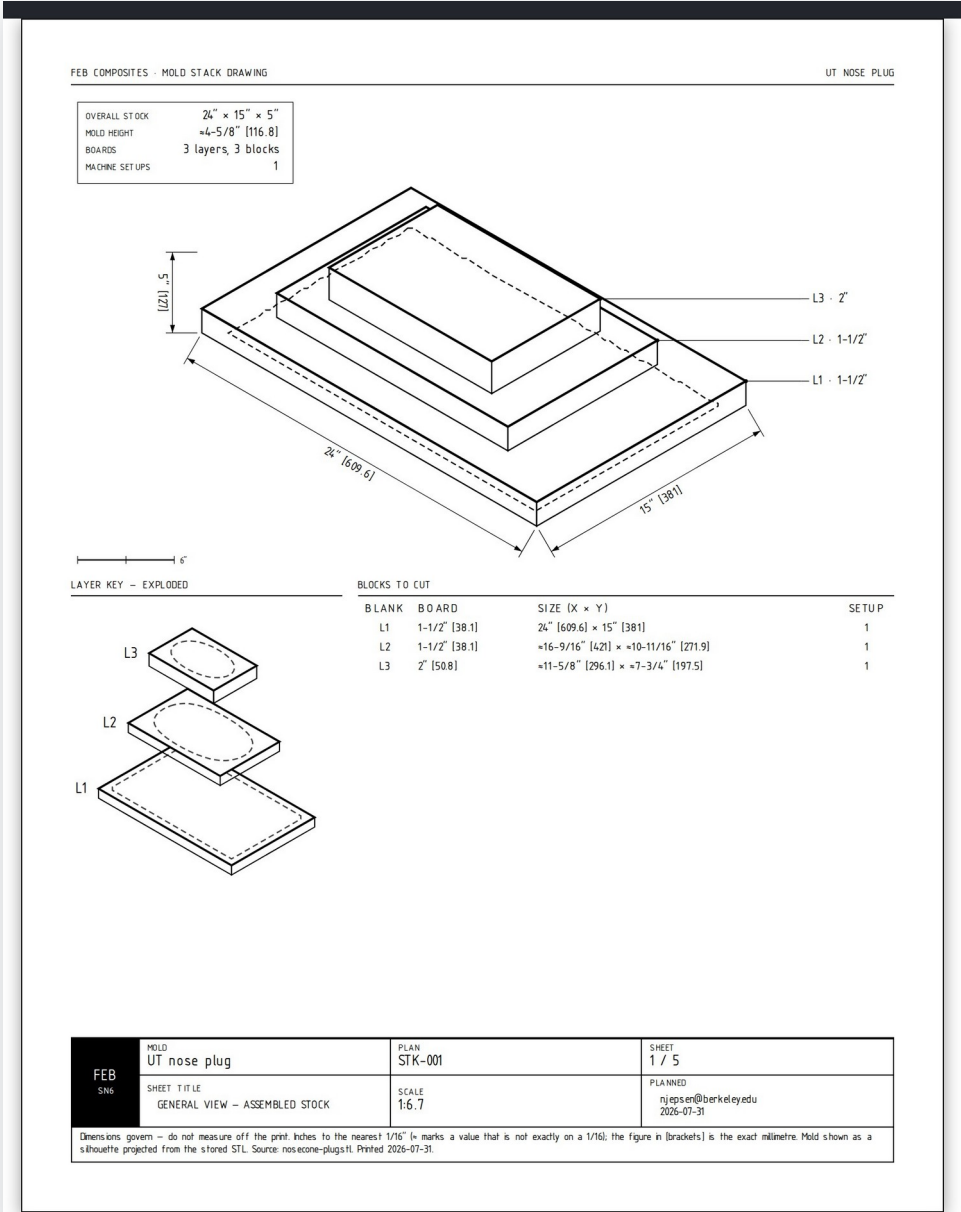
The blocks come back out as CAM stock

Export stock STL writes one file per machine setup, in millimetres, at the mold's own CAD origin. It drops onto the model in CAD with nothing to align, so CAM uses it as the stock body directly. Marked beta.

The boards get glued by hand, so it prints a dimensioned sheet per layer

Whoever is holding layer 3 needs to know how far in from each edge of layer 2 it goes.

Inches to the nearest 1/16in with the exact millimetre bracketed. Every sheet marks the same datum corner.



The traveler carries the stack, the BOM and the standard each step answers to

Stack frozen at a revision — any change past that point needs a new one.

Every step cites the CS clause it comes from.

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Restore

Load SN5 archive

Roster

Nick Jepsen · lead

Sign out

< All work orders

Edit

Print

Create issue

Overview

Stack

BOM

Steps

Quality

Log

WO-SN6-004 · UT NOSE

Rev B · In Work

OVERVIEW

PART NAME	SUBTEAM	STATUS	PROCESS	MOLD ENGINEER
UT NOSE	Aero	In Work	MoldInfusion	Ana Rivera
MANUFACTURING ENGINEER	CREATED	DUE	REVISION	MASS TARGET (G)
David Okafor	2026-07-20	2026-08-14	B	740
MASS ACTUAL (G)				
—				
MOLD				
MOLD ID	LAYERS	DENSITY (LB/FT³)	SEALING	LOCATION (UPDATE ON EVERY MOVE)
STK-001	3	30	XCR, 2 coats	RFS — rack B, bay 2

LAYUP STACK · Frozen 2026-07-24. Any change past this point needs a new revision.

P1 · 195 twill · 0/90 · full

P2 · 88 spread-tow · +45 · full

P3 · Nomex honeycomb 0.125" — core, 40mm inset · chamfer the edge

P4 · 88 spread-tow · -45 · full

P5 · 195 twill · 0/90 · full

BOM

ITEM	QTY	UNIT	SOURCE	EST. COST
195 twill 2×2, 3k	2.4	yd	Sigmatex	168.00
88 spread-tow	1.8	yd	Sigmatex	121.00
Nomex honeycomb 0.125"	0.5	sheet	ACP	46.50

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Work Orders

OverviewStackBOMStepsQualityLog

STEPS AND BUY-OFFS (BLOCKERS SHADED: NO SIGN-OFF, NO MOVING ON)

1Stack frozenBLOCKER✓ Ana Rivera 2026-07-24

2Mold design reviewBLOCKER✓ Nick Jepsen 2026-07-27

3Seal and release the mold✓ Mei Chen 2026-07-29

4Dry stack + bagbuy off as Nick Jepsen

5Drop testBLOCKERbuy off as Nick Jepsen

6Infusebuy off as Nick Jepsen

7Cure + demouldbuy off as Nick Jepsen

8Trim and finishbuy off as Nick Jepsen

QUALITY CHECKS / ACCEPTANCE CRITERIA

CRITERION	TARGET (SET AT CREATION!)	ACTUAL	PASS
mass	740		—
no dry spots at the corner radius	none visible		—
bag holds 25 inHg for 10 min	≤ 1 inHg drop	0.5 inHg	pass

EVENT LOG

DATE	EVENT
2026-07-24	Stack frozen at Rev B after the core inset changed
2026-07-27	Mold design review signed — fit checked against STK-001
2026-07-29	Two XCR coats, buffed between

NOTES

First SN6 part through the planner end to end: STK-001 cut list, then this traveler.

04 THE TRAVELER

Buy-offs carry a name, and blockers hold the line

Stack freeze, mold design review and drop test are blockers: later steps stay locked until they're signed. It records who was signed in. It does not prove who held the part, and a roster member can still edit the record afterwards.

It measures the sheet before it prints it, to hold a two-page traveler

The app renders the sheet, measures what the browser actually laid out, and steps down through tighter layouts until it fits.

Blank travelers for the bench. Always black-on-white, whatever theme you're in.

FEB COMPOSITES 5100

WORK ORDER
WO-SN6-004

REV
B

PROCESS
MOLD INFUSION

MANUFACTURING TRAVELER. FILL IT IN AT THE BENCH, THEN TRANSCRIBE IT INTO THE APP. PRINTED 2026-07-31

PART AND ASSIGNMENT

PART NAME UT NOSE		SUBTEAM Aero	STATUS In Work
MOLD ENGINEER Ana Rivera	MANUFACTURING ENGINEER David Okadoi	CREATED 2026-07-20	DUE 2026-08-14
MASS TARGET (G) 740	MASS ACTUAL (G)	MOLD ID STK-001	MOLD LOCATION RFS --- rack B, bay 2

MOLD

TOOLING BOARD LAYERS 3	DENSITY (LB/FT) 30	SEALING SYSTEM XCR, 2 coats
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LAYUP STACK

Frozen 2026-07-24. Any change past this point needs a new revision.

PLY	MATERIAL	ORIENTATION	COVERAGE	NOTES
1	195 twill	0/90	full	
2	88 spread-tow	+45	full	
3	Nomex honeycomb 0.125"	---	core, 40mm inset	chamfer the edge
4	88 spread-tow	-45	full	
5	195 twill	0/90	full	

STEPS AND BUY-OFFS

Initial and date each step as it is completed

#	OPERATION	INITIAL	DATE
1	☐ Stack frozen ☒ Blocker: no sign-off, no moving on	Ana Rivera	2026-07-24
2	☐ Mold design review ☒ Blocker: no sign-off, no moving on	Nick Jeppsen	2026-07-27
3	☐ Seal and release the mold	Mei Chen	2026-07-29
4	☐ Dry stack + bag		/ /
5	☐ Drop test ☒ Blocker: no sign-off, no moving on		/ /
6	☐ Infuse		/ /
7	☐ Cure + demould		/ /
8	☐ Trim and finish		/ /

BILL OF MATERIALS

ITEM	QTY	UNIT	SOURCE	EST. COST
195 twill 2x2, 3k	2.4	yd	Signatex	168.00
88 spread-tow	1.8	yd	Signatex	121.00
Nomex honeycomb 0.125"	0.5	sheet	ACP	46.50
IN2 infusion resin + AT30	1.2	kg	Easy Composites	52.80
Peel ply / flow mesh / bag film	1	set	Airtex	38.20

QUALITY CHECKS

Set the target before the work starts

CRITERION	TARGET	ACTUAL	PASS
mass	740		
no dry spots at the corner radius	none visible		
bag holds 25 inHg for 10 min	≤ 1 inHg drop	0.5 inHg	PASS

EVENT LOG

What actually happened, including what went wrong

DATE	EVENT
2026-07-24	Stack frozen at Rev B after the core inset changed
2026-07-27	Mold design review signed --- fit checked against STK-001
2026-07-29	Two XCR coats, buffed between
/ /	

NOTES

First SN6 part through the planner end to end: STK-001 cut list, then this traveler.

RELEASE SIGN-OFF

MANUFACTURING ENGINEER David Okadoi	COMPOSITES LEAD OR REQUESTING SUBTEAM LEAD
Signature _____ Date _____	Signature _____ Date _____

WO-SN6-004 - REV B - UT NOSE

PAGE _____ OF _____

Everything that isn't a part lives in Tickets

R&D, process fixes, bugs, outreach.
Board or list, assignees, watchers,
sub-tickets.

Cross-linked to the parts and work
orders they touch.

The screenshot displays the FEB Composites Tickets interface. On the left is a dark blue sidebar with navigation icons and labels: Dashboard, Work Orders, Parts, Stock, Tickets (highlighted), Timeline, Weekly Plan, Budget, Documents, Reports, and People. The main area is titled 'Tickets' and features a Kanban board with columns: TO DO (4 tickets), IN PROGRESS (3 tickets), COLLECTING DATA (0 tickets), ON HOLD (1 ticket), DONE (1 ticket), and CANCELLED (0 tickets). Each ticket card includes a 'PROJECT' label, a title, a priority (High, Medium, Low), a due date, and an assignee's initials in a colored circle. For example, the first 'TO DO' ticket is 'Order carbon before the customs window closes' (High priority, due 2026-08-04, assigned to NJ). The 'ON HOLD' ticket is 'Drop-test rig needs a new load cell' (High priority, due 2026-07-24, assigned to TB). The top right of the interface has a search bar, a bell icon, a refresh icon, and buttons for Backup, Restore, Load SN5 archive, Roster, and a user profile for Nick Jepsen (lead) with a Sign out button.

The weekly plan pulls each person's tasks from their ticket due dates

Plus carpools — who's driving to RFS, when, and how many seats are left.

Logistics stop living in one person's head, or in a thread that scrolls away.

The screenshot displays the 'Weekly Plan' interface for 'FEB Composites'. The left sidebar contains navigation links: Dashboard, Work Orders, Parts, Stock, Tickets, Timeline, Weekly Plan (highlighted), Budget, Documents, Reports, and People. The main content area is titled 'Weekly Plan' and shows a dropdown for '2026-07-27 (W00) · this week'. Below this, a section titled 'WEEKLY GOALS — WEEK OF 2026-07-27' lists tasks for six team members, each with a status indicator (checkbox or checkmark), a due date, and a trash icon.

Name	Task	Status	Due Date
Nick Jepsen (you)	Sign the CS-003 approval table	☐	due 2026-07-31
Ana Rivera	Seal and wax the nose plug — two XGR coats	☒	2d late
Ana Rivera	Cut the 45° plies for the UT skin	☐	due 2026-07-31
David Okafor	Re-route the infusion spiral at the corner	☐	1d late
Jai Patel	Leak-check the bag before Thursday's infusion	☐	1d late
Lena Novak	Draft the Sigmatex follow-up email	☒	2d late
Mei Chen	Photograph the rack and reconcile the offcuts	☒	3d late

Purchases run Submitted, Ordered, Reimbursed

Anything over \$50 is flagged for approval, per CS-012.

SN5 ran about \$5.3k through one person's personal card — the totals on this screen are demo data.

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\$2054

SEASON TOTAL

5

OPEN ORDERS (\$1616)

2

OVER \$50, UNAPPROVED

+ New Purchase

All statuses

search item / purchaser...

ITEM	PURCHASER	PURPOSE	STATUS	COST	ORDERED
Load cell, 500 lbf	Jai Patel	Testing	Submitted	\$189.00	2026-09-21
Peel ply, 5 yd	David Okafor	Restock	Submitted	\$44.15	2026-09-17
Vacuum bag film, 60in x 25 yd	Simon Starbuck	Restock	Submitted	\$132.90	2026-09-13
Tooling board 40 lb/ft3, 2in x 24in x 48in	Mei Chen	Tooling	Ordered	\$412.00	2026-09-09
IN2 infusion resin, 5 kg + AT30 hardener	Ana Rivera	Manufacturing	Reimbursed	\$214.60	2026-09-05
Mixing cups and sticks, bulk	Mei Chen	Other	Reimbursed	\$27.30	2026-08-19
Spiral wrap tubing, 50 ft	Ana Rivera	Restock	Reimbursed	\$31.80	2026-08-15
Breather cloth, 10 yd	Jai Patel	Manufacturing	Reimbursed	\$96.25	2026-08-11
XCR mold sealer, 1 qt	David Okafor	Tooling	Reimbursed	\$68.40	2026-08-07
Carbon twill 2x2, 3k, 50in — 12 yd	Simon Starbuck	Manufacturing	Ordered	\$838.00	2026-08-03

Backup

Restore

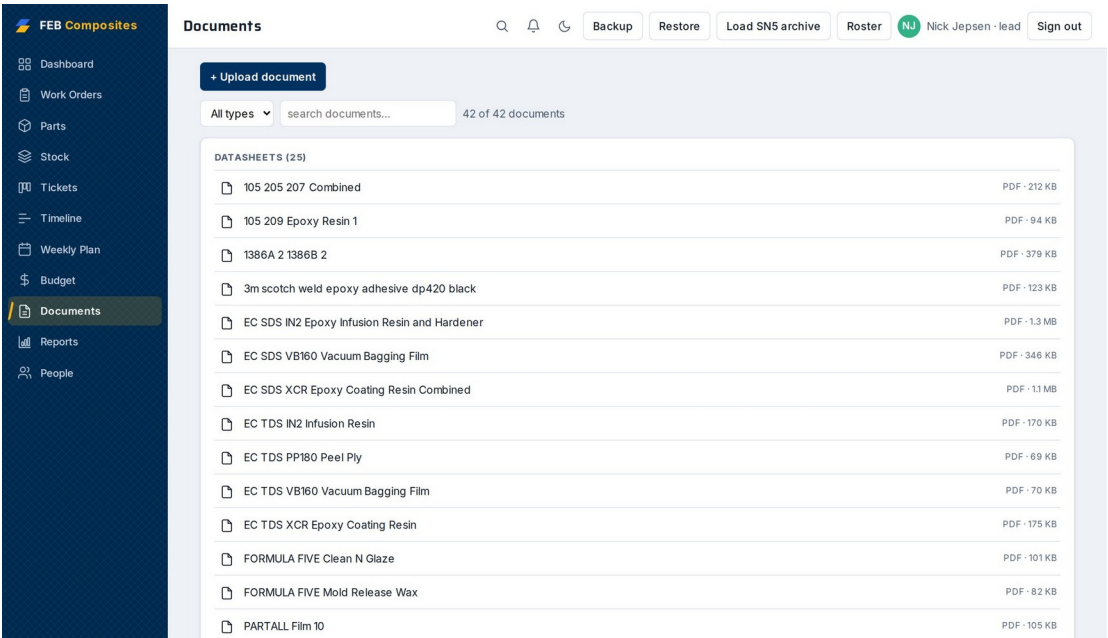
Load SN5 archive

Roster

NJ Nick Jepsen · lead

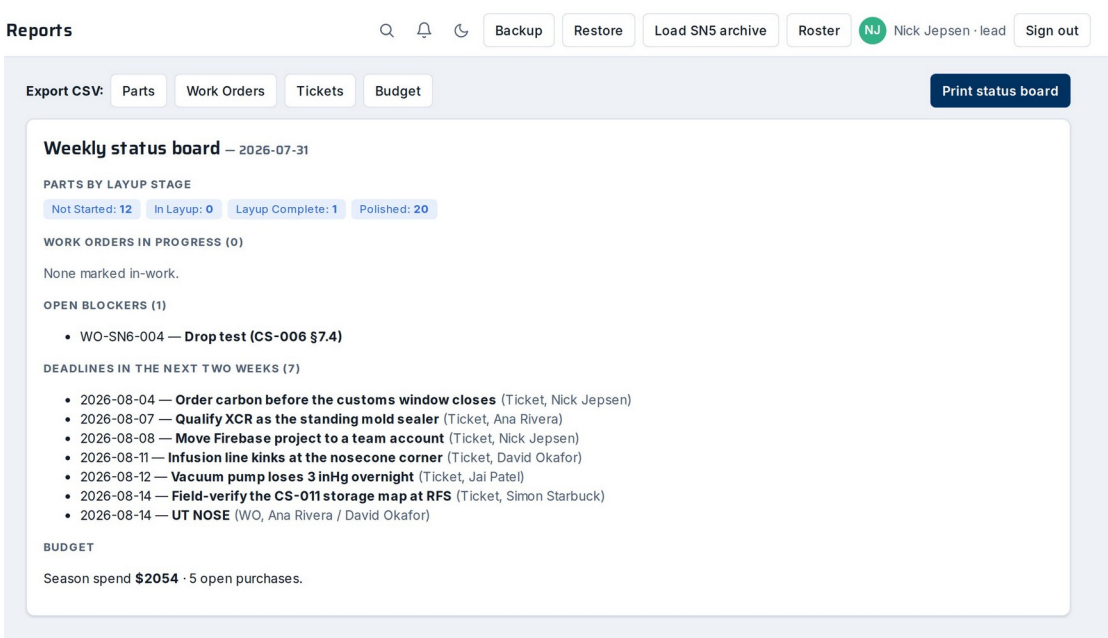
Sign out

Every reference doc in the app



Documents

25 manufacturer datasheets, the CS standards rendered from source, and the shop printables. Filterable. Anyone can upload.



Reports

CSV export per dataset, plus the Monday-meeting status board as a single printable page.

Built to still run in two years

0 build steps

No framework, no bundler — 24 plain script tags. Open the file and it runs.

1 dependency

DOMPurify, SRI-pinned and self-hosted after shop wifi silently degraded it from a CDN.

8,900 lines of app source

Small enough that the next lead can read it. The 3D viewer is hand-rolled WebGL with nothing behind it.

Two of the test suites render the real thing in a browser and measure it: every drawing sheet across 8 mold fixtures, checked for legibility down to a 5.5pt floor, and the whole app booted at 4 device widths with every printable opened.

What it does not do

Buy-offs are not tamper-proof

They record who was signed in. Any roster member can still edit any record, and there is no in-app version history. Monthly JSON exports to Drive are the audit trail.

Notifications are in-app only

Watcher unread state lives in one browser. It does not sync between devices and it does not email.

Four standards are outlined, not drafted

CS-001, CS-007, CS-008 and CS-009. Outlined means the structure is there and the content isn't.

STL export is beta

Real exports still turn up surprises. Check the stack view before anyone cuts.

Four decisions

WHO

01	Adopt it for SN6	Roster everyone in week 1, so it is the one place the season is written down.	Week 1
02	Move the Firebase project	feb-composites sits on a personal Google account. It has to move to a team account, or it is one graduation away from gone.	Before SN6 kickoff
03	Sign the CS approval tables	All 14 standards ship “Draft, pending Lead signature”.	Sept
04	Field-verify CS-011 at RFS	The storage map was written from records, not from standing in the room.	Sept